

Customer Retention & Churn Analysis

Statistical Analysis with Business Recommendations

Future Interns

Data Science & Analytics - Task 2
Customer Retention & Churn Analysis

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Date: June 28, 2026

Repository: FUTURE_DS_02

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1 Executive Summary

This report analyzes **7,043 customers** from a telecommunications company to identify churn patterns, key risk factors, and retention strategies. Using chi-square tests and logistic regression, we identify the strongest predictors of customer churn and provide evidence-based recommendations.

1.1 Key Performance Indicators

Metric	Value
Total Customers	7,043
Churn Rate	26.54%
Churned Customers	1,869
Retained Customers	5,174
Avg Monthly Charge	\$64.76
Monthly Revenue at Risk	\$121,040
Yearly Revenue at Risk	\$1,452,475

Table 1: Churn Key Performance Indicators

Critical Finding: Over 1 in 4 customers churn, putting **\$1.45 million** in annual revenue at risk. Month-to-month contracts (42.7% churn), Fiber Optic service (41.9%), and Electronic Check payments (45.3%) are the strongest risk factors identified.

2 Data Overview

Attribute	Description
Dataset Source	Telco Customer Churn Dataset
Total Customers	7,043
Features	21 (demographics, services, billing)
Target Variable	Churn (Yes/No)
Data Quality	11 missing TotalCharges (fixed)
Duplicate Records	0

Table 2: Dataset Characteristics

2.1 Business Questions Addressed

Q1: Why are customers leaving?

Q2: Which segments are most at risk?

Q3: How long do customers typically stay?

Q4: What services correlate with retention?

Q5: What actions can reduce churn?

3 Churn Analysis Results

3.1 Churn by Contract Type

Contract Type	Customers	Churned	Churn Rate
Month-to-Month	3,875	1,655	42.71%
One Year	1,473	166	11.27%
Two Year	1,695	48	2.83%

Table 3: Churn Rate by Contract Type

Key Finding: Contract Type is the #1 Predictor

Month-to-month customers are **15 times more likely to churn** than two-year contract customers (42.71% vs 2.83%). Converting month-to-month customers to annual contracts represents the single biggest retention opportunity.

3.2 Churn by Customer Tenure

Tenure Group	Customers	Churned	Churn Rate
0–12 months	2,186	1,037	47.44%
13–24 months	1,024	294	28.71%
25–36 months	832	180	21.63%
37–48 months	762	145	19.03%
49–60 months	832	120	14.42%
60+ months	1,407	93	6.61%

Table 4: Churn Rate by Customer Tenure

3.3 Churn by Internet Service

Internet Service	Customers	Churned	Churn Rate
Fiber Optic	3,096	1,297	41.89%
DSL	2,421	459	18.96%
No Internet	1,526	113	7.40%

Table 5: Churn Rate by Internet Service Type

3.4 Churn by Payment Method

Payment Method	Customers	Churned	Churn Rate
Electronic Check	2,365	1,071	45.29%
Mailed Check	1,612	308	19.11%
Bank Transfer (Auto)	1,544	258	16.71%
Credit Card (Auto)	1,522	232	15.24%

Table 6: Churn Rate by Payment Method

3.5 Churn by Senior Citizen Status

Senior Citizen	Customers	Churned	Churn Rate
No	5,901	1,393	23.61%
Yes	1,142	476	41.68%

Table 7: Churn Rate by Senior Citizen Status

4 Statistical Analysis

4.1 Chi-Square Tests - Identifying Significant Factors

17 out of 19 variables tested showed a statistically significant relationship with churn ($p < 0.05$). Only **Gender** and **Phone Service** were not significant predictors.

4.2 Cramer's V - Effect Size

Variable	Cramer's V	Strength
Contract Type	0.410	Moderate
Tenure Group	0.352	Moderate
Online Security	0.347	Moderate
Tech Support	0.343	Moderate
Internet Service	0.322	Moderate
Payment Method	0.303	Moderate
Online Backup	0.292	Weak
Device Protection	0.282	Weak
Monthly Charges Group	0.234	Weak
Streaming Movies	0.231	Weak
Streaming TV	0.231	Weak
Has Internet	0.228	Weak
Paperless Billing	0.192	Weak
Dependents	0.164	Weak
Senior Citizen	0.151	Weak
Partner	0.150	Weak
Multiple Lines	0.040	Negligible
Phone Service	0.011	Negligible
Gender	0.008	Negligible

Table 8: Cramer's V Effect Size for All Variables

4.3 Logistic Regression - Quantifying Risk

Model: Churn $\sim$ Contract + Tenure + Internet + Payment + Services + Senior Citizen + Paperless Billing

Metric	Value
Pseudo R-squared	0.2689
Accuracy	79.81%
Specificity	90.34%
Sensitivity	50.67%
AIC	5,990.85

Table 9: Logistic Regression Model Performance

4.4 Odds Ratios - What Drives Churn?

4.4.1 Factors That INCREASE Churn Risk

Factor	Odds Ratio	Interpretation
Fiber Optic Internet	2.70×	170% higher churn risk
Electronic Check Payment	1.52×	52% higher churn risk
Paperless Billing	1.48×	48% higher churn risk
Senior Citizen	1.37×	37% higher churn risk

Table 10: Top Churn Risk Increaseers

4.4.2 Factors That DECREASE Churn Risk

Factor	Odds Ratio	Interpretation
Two-Year Contract	0.17×	83% lower churn risk
60+ Months Tenure	0.19×	81% lower churn risk
One-Year Contract	0.44×	56% lower churn risk
No Internet Service	0.48×	52% lower churn risk

Table 11: Top Churn Risk Reducers

Statistical Conclusion

The strongest actionable finding: **Moving a customer from month-to-month to a two-year contract reduces their churn odds by 83%.** This single intervention could retain hundreds of customers and save over \$1M in annual revenue.

5 Recommendations

5.1 Immediate Actions (0–3 Months)

- 1. Launch contract conversion campaign** - Target month-to-month customers with incentives to switch to annual/two-year contracts
- 2. Address Fiber Optic churn** - Investigate service quality issues; 41.9% churn suggests product problems
- 3. Promote automatic payments** - Electronic check users have 45.3% churn vs 15.2% for credit card auto-pay

5.2 Strategic Initiatives (3–12 Months)

- 4. First-year retention program** - 47.4% churn in first 12 months requires onboarding improvements
- 5. Senior citizen support** - 41.7% churn rate; develop specialized retention team
- 6. Service bundling strategy** - Customers with 7+ services have only 8.7% churn

5.3 Estimated Financial Impact

Reducing churn by just **5 percentage points** (from 26.5% to 21.5%) would retain approximately **350 additional customers** and save **\$272,000 in annual revenue**.

6 Methodology

Technique	Purpose
Data Cleaning	Handle missing values, create derived features
Descriptive Statistics	Churn rate by segment, KPI calculation
Chi-Square Tests	Identify statistically significant churn factors
Cramer's V	Measure effect size of each predictor
Logistic Regression	Quantify churn risk (odds ratios)
Confusion Matrix	Evaluate model performance

Table 12: Analytical Methodology

Tools Used: R Statistical Computing Environment, tidyverse, broom, ggplot2

7 Conclusion

This analysis identified **26.54% churn rate** representing **\$1.45M in annual revenue at risk**. Statistical testing confirmed that **17 of 19 variables** significantly predict churn.

The Churn Profile: The highest-risk customer is on a **month-to-month contract**, paying by **electronic check**, using **Fiber Optic internet**, and in their **first year** of service. This customer is **15× more likely to churn** than a two-year contract customer.

Key Takeaways:

- **Contract type** is the single strongest predictor – two-year contracts reduce churn by 83%
- **First-year retention** is critical – 47.4% of new customers leave within 12 months
- **Payment method matters** – automatic payments show significantly lower churn
- **Service bundling works** – 7+ services reduces churn to single digits
- **Fiber Optic needs investigation** – 2.7× churn risk suggests product issues

The recommendations in this report provide a clear, evidence-based roadmap for reducing churn and protecting over \$1.4M in annual revenue.

Reducing churn by 5 points = \$272,000 in saved annual revenue



Report prepared for **Future Interns - Data Science & Analytics Task 2**

Analysis conducted using R Statistical Computing Environment

June 28, 2026